



Land use & land cover dynamics and its relationship with nitrate pollution in groundwater around inactive mine areas using geospatial techniques, SW part of Cuddapah basin, Southern India

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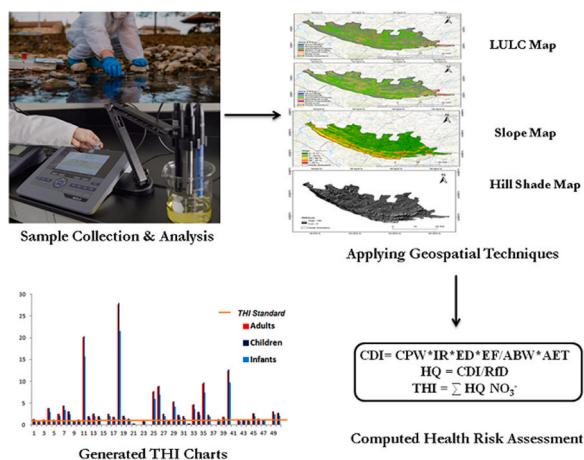
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HIGHLIGHTS

- 32% and 14% of the samples fall in high risk zones of NO_3^- concentration levels.
- LULC reports increase in cropland cause NO_3^- pollution prevalent in irrigated areas.
- Impoundments are found in slope areas lead to added NO_3^- pollutants in to aquifers.
- From THI values 34% of the children maybe susceptible to blue baby syndrome disease.

GRAPHICAL ABSTRACT



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ABSTRACT

Geospatial maps can show how the ineffective operations of inactive mines affect water and aquifer quality. As such, the purpose of this study is to assess the impact of mining and irrigation on the aquifer ecosystem through the evaluation of LULC and slope maps through the application of Landsat 8 OLI/TIRS and DEM data. A total of 50 groundwater samples were prepared from villages in the close proximity to inactive mines during pre and post monsoon periods in 2021. The results of the analysis revealed alarming statistics, that 14% of groundwater samples exceeded the WHO nitrate limit in pre & post monsoon season, indicating a high-risk in the study area. According to guidelines (USEPA, 2014), 34% in pre-monsoon and 26% post-monsoon of samples exceeded the THI levels for adults and children respectively, indicating non-carcinogenic health risks. In addition, 80% of the samples in both seasons exhibited high NPI values, indicating nitrate contamination associated with blue baby syndrome. From the Geospatial analysis the findings from the LULC classification indicate that there has been a

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